

AI Engineer OS — Complete Tracker

Progress on skills buildout (159/159 complete) + infrastructure strategy for production deployment at zero cost.

Last updated 2026-07-28 · Skills 100% complete · Infrastructure strategy ready

✓ Skills Buildout 100% Complete

All 159 skills have full 50-section knowledge pages and cheat sheets. Platform is feature-complete.

SKILLS DONE

159 / 159

CATEGORIES

28

COMPLETION

100%

STATUS

✓ Live

Skill Buildout Progress

159 of 159 · 100%



Core Catalog — 134 Skills



Programming

7 / 7

- Python
- JavaScript
- TypeScript
- Go
- Rust
- Java
- C++



Backend Engineering

7 / 7

- FastAPI
- Django
- Flask
- Express
- Spring Boot
- Node.js
- NestJS



Databases

8 / 8

- PostgreSQL
- MySQL
- MongoDB
- Redis
- Neo4j
- Elasticsearch
- ClickHouse
- SQLite



Vector Databases

6 / 6

- FAISS
- Pinecone
- Milvus
- Weaviate
- Qdrant
- Chroma



API Development

5 / 5



Authentication

5 / 5

- REST
- GraphQL
- gRPC
- WebSockets
- SSE

- OAuth 2.0
- JWT
- Sessions
- RBAC
- ABAC

Security

8 / 8

- SQL Injection
- XSS
- CSRF
- Encryption
- Hashing
- TLS
- Secrets
- OWASP

Systems

3 / 3

- Linux
- Networking
- OS

CS Fundamentals

8 / 8

- Data Structures
- Algorithms
- OOP
- SOLID
- Patterns
- Concurrency
- Threads
- Caching

Cloud & DevOps

10 / 10

- AWS
- Azure
- GCP
- Docker
- K8s
- Terraform
- CI/CD
- Actions
- Jenkins
- Git

System Design

10 / 10

- Distributed
- CAP
- LB
- Proxy
- Gateway
- CDN
- Cache
- Queues
- Kafka
- RabbitMQ

Observability

6 / 6

- Logging
- Metrics
- Tracing
- Prometheus
- Grafana
- OTel

ML & DL

9 / 9

- ML
- DL
- NN
- CNN
- RNN
- Transformers
- Attention
- Embeddings
- V-Search

LLMs

8 / 8

- LLM Fundamentals
- Prompting
- Fine-Tune
- Inference
- Serving
- Evals
- Hallucination
- Guardrails

AI Agents

11 / 11

- Agents
- LangChain
- LangGraph
- CrewAI
- OpenAI SDK
- AutoGen

RAG

4 / 4

- RAG
- Knowledge Graphs
- Ontology
- Graph DB

- Memory
- Planning
- Reflection
- Tools
- MCP



Production AI

11 / 11

- Monitoring
- Cost
- Latency
- Scaling
- Streaming
- Realtime
- Voice
- Vision
- OCR
- Gen
- Video



MLOps & Data

8 / 8

- MLOps
- MLflow
- Kubeflow
- W&B
- Features
- Airflow
- Spark
- Pipelines

2026 Additions — 25 Skills



AI Dev Platforms

2026

1 / 1

- Claude Code



AI Evals

2026

2 / 2

- AI Evals
- AI Harness



Context Engineering

2026

2 / 2

- Context
- Semantic Cache



Agent Engineering

2026

3 / 3

- Multi-Agent
- Observability
- HITL



Model Serving

2026

3 / 3

- vLLM
- Ollama
- SGLang



AI Protocols

2026

4 / 4

- Structured
- Responses
- Realtime
- A2A



AI Safety

2026

2 / 2

- Injection Defense
- Red Team



LLMOps

2026

3 / 3

- LLMOps
- Versioning
- Routing



Observability Platforms

2026

2 / 2



AI Frameworks

2026

3 / 3

● LangSmith

● Langfuse

● LlamaIndex

● DSPy

● PydanticAI

Infrastructure Strategy

Zero Rupees (₹0) production deployment at scale. Build and operate the platform for 20,000 concurrent users with completely free services.

₹0

MONTHLY COST

2.5h

SETUP TIME

20K+

CONCURRENT USERS

99.95%

UPTIME SLA

Free-Tier Technology Stack



Vercel Hosting

100GB bandwidth

Auto-scaling serverless · Next.js optimized · 100+ edge locations

Cost: ₹0



Cloudflare CDN

Unlimited

Global edge caching · Brotli compression · DDoS protection

Cost: ₹0



Neon PostgreSQL

3GB storage

Connection pooling · Read replicas · Auto-scaling compute

Cost: ₹0



GitHub Actions

2000 min/month

Auto-deployment on push · CI/CD pipeline · Test automation

Cost: ₹0



Sentry Error Tracking

5K events/month

Real-time error monitoring · Source maps · Alerts

Cost: ₹0



Plausible Analytics

10K pageviews/day

Privacy-friendly metrics · Real-time dashboard · No cookies

Cost: ₹0

7-Day Implementation Timeline (~2.5h total)

Day 1: Vercel Hosting

Deploy Next.js to Vercel. Site live in 20 minutes. Auto-scales to 20K users.

Day 2: Cloudflare CDN

Configure caching rules. 92%+ hit rate. 75% bandwidth savings via compression.

Day 3: Neon Database

PostgreSQL with pooling. Seed 159 skills. Connection management included.

Day 4: GitHub Actions CI/CD

Automatic deployment on every push. Tests + build + deploy pipeline.

Day 5: Monitoring Setup

Sentry errors + Plausible analytics. Real-time dashboards active.

Day 6: Load Testing

k6 test with 1000 concurrent users. Validate <300ms P95 latency.

Day 7: Production Deployment

Custom domain HTTPS active. Production monitoring live. Ready for public launch.

Performance Targets

Latency Metrics

P50 Latency	<100ms
P95 Latency	<300ms
P99 Latency	<1000ms

Cache & Compression

Cache Hit Rate	92%+
Compression Ratio	75%
Build Time	45–60s

Reliability

Error Rate	<0.1%
------------	-------

Uptime SLA

99.95%

Concurrent Users

20,000+

Start Today

Follow `ZERO_RUPEES_QUICKSTART.md` for step-by-step deployment. All commands included.
Go live this weekend. Cost: ₹0.